

Solid-state poly(methyl methacrylate) (PMMA) nanofoams. Part I: Low-temperature CO₂ sorption, diffusion, and the depression in PMMA glass transition



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ABSTRACT

In this paper, we report on the solubility and diffusivity of CO₂ in PMMA in the range -30°C to 100°C , and on the depression in the glass transition temperature when the polymer acquires an equilibrium mass% of CO₂ in this temperature range. Solubility of CO₂ in PMMA increases ten-fold from 3.9% at 100°C to 39.3% at -30°C . We observe a drop in ΔH_s , the heat of sorption, at the transition of CO₂ from vapor phase to liquid phase, leading to a weaker dependence of solubility on temperature in the liquid CO₂ regime. Four distinct diffusivity regions are identified, where diffusivity may increase or decrease with the decreasing temperature, characterized by the different activation energies. Besides the two transitions (normal vitrification and retrograde vitrification) between glassy state and rubbery state observed previously in PMMA-CO₂, we also find a new transition in the vicinity of CO₂ vaporization temperature. The dissolution of CO₂ in PMMA lowers its glass transition temperature in a non-linear fashion and the incorporation of 39.3% CO₂ in PMMA decreases its T_g from 103°C to -12.5°C . In part II of this paper, we describe a process, based on low-temperature CO₂ saturation, that creates PMMA nanofoams with 30–40 nm cells.

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1. Introduction

Nanofoams are thermoplastics with nano size bubbles. In the past three decades, a number of gas-polymer systems have been studied, and foams with cell size of 100 nm and higher have been produced (see Part II of this paper for a brief review of solid-state microcellular and nanocellular foams).

The main approach to reducing cell size is based on increasing the cell nucleation density. It is well known that a higher amount of gas absorbed in the polymer leads to a higher number of cells to nucleate in a unit volume of polymer [1,2]. Thus higher CO₂ pressures have been used to reduce cell size from say $1\text{ }\mu\text{m}$ – $0.1\text{ }\mu\text{m}$ (or 100 nm). It is also known that the gas solubility in polymer increases, in general, as the saturation temperature decreases [3]. This research was undertaken to explore if the higher gas solubility at lower saturation temperature can be used to in a solid-state process [1,2,4,5] to drive down the cell size from order 100 nm to

order 10 nm. This paper addresses the solubility and diffusivity of CO₂ in PMMA at low temperatures, up to -30°C , as well as the effect of absorbed CO₂ on the glass transition in PMMA. In Part II, we explore the process-space for solid-state foaming in PMMA-CO₂ at low temperatures, and present a processing window in which PMMA nanofoams with cells in the 30–40 nm range can be produced.

There are a number of studies on the PMMA-CO₂ system in the literature. Wissinger and Paulaitis studied sorption of CO₂ in PMMA at 32 – 65°C and pressures up to 10 MPa [6,7]. Kamiya et al. investigated CO₂ sorption in PMMA in the range of 35 – 200°C and pressures up to 6 MPa [8]. Shieh and Liu studied sorption and CO₂/PMMA interactions at 32°C and pressures up to 35 MPa [9]. Rajendran et al. investigate the sorption behavior of CO₂ in PMMA at 50, 65, and 80°C , and pressures up to 23.8 MPa [10]. Vogt et al. studied sorption of CO₂ in PMMA 60 and 100°C , and pressures up to 12 MPa [11].

PMMA-CO₂ is known to exhibit the retrograde vitrification behavior [12–16]. At a given gas pressure, two T_g s are observed. One of the T_g s is the normally expected plasticized T_g corresponding to the glass-to-rubber transition. In addition, another T_g is

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observed at the same pressure, but at a lower temperature, and corresponds to the rubber-to-glass transition upon heating. This unusual behavior observed has been termed retrograde vitrification [17]. The retrograde behavior of PMMA has been utilized to create ultramicrocellular foams in the solid-state foaming process. PMMA foams with an average cell size of 350 nm, a cell density of 4.4×10^{13} cells/cm³ and a relative density of about 10%, were obtained by saturating in CO₂ at -0.2 °C and 3.4 MPa [16]. One advantage of using the retrograde vitrification behavior is that gas sorption can occur much faster when the polymer-gas system is in the rubbery phase, for example, at 3.4 MPa, gas sorption occurs much more rapidly at -0.2 °C than at 24 °C¹⁶. The other advantage is that only low pressure is required for producing foams with similar or finer microstructures [18].

In this paper, we first discuss the effect of temperature on sorption behavior of CO₂ in the range of -30 °C up to 100 °C. Both solubility and diffusivity are studied. Arrhenius relationship is used to describe the effect of temperature on these parameters. Finally, glass transition temperatures of PMMA–CO₂ mixture are estimated and compared to commonly used models.

2. Experimental

2.1. Material

Acrylite FF PMMA sheets manufactured by CYRO Industries (New Jersey, USA) with a thickness of 1.48 mm were purchased. Sheets were cut into 2.5 cm × 2.5 cm samples using a band saw. The PMMA has a $M_w = 134,000$ g/mol, $M_n = 71,000$ g/mol, and a density of 1.19 g/cm³. Glass transition temperature (T_g) was measured to be 103 °C in differential scanning calorimeter (DSC) TA Instruments Q20, with a heating rate of 10 °C/min. T_g was determined using the half-height method. Medical grade CO₂ (99.9% purity) was purchased from Praxair, Inc.

2.2. Sorption

Sorption experiments were conducted by placing samples in a pressure vessel, with the CO₂ pressure inside maintained at 5 MPa with an accuracy of ± 0.1 MPa. In order to maximize CO₂ sorption in polymer and eliminate the need of a pressure boosting device, we chose 5 MPa as the saturation pressure, which was slightly below the pressure of a full CO₂ gas cylinder at room temperature. Saturation temperature (T_{sat}) varied over a wide range from -30 °C to 100 °C. For sorption experiment above room temperature, a heating jacket wrapped around the pressure vessel and a temperature controller was used to maintain the pressure vessel at a desired temperature. For the temperatures between room temperature and 0 °C, a Peltier cooling module was used. For temperatures below 0 °C, the pressure vessel was placed in a freezer capable of achieving -30 °C to 0 °C. During saturation, samples were periodically taken out from the pressure vessel, and weighed on a Mettler AE240 analytical scale accurate to ± 10 µg. Samples were then promptly put back to the pressure vessel and repressurized. The sorption experiment continued until no further weight increase was observed in the specimen.

2.3. Glass transition temperature determination

Glass transition temperature was determined by a minimum foaming temperature approach. Specimens were first placed in a pressure vessel and then allowed to absorb CO₂ over a pre-determined amount of time (based on sorption studies). After full saturation, samples were removed from the pressure vessel, and immediately immersed in a silicone oil bath (Thermo Haake B5) set

at a foaming temperature (T_f) in the range of -20 °C to 120 °C. The samples were kept in the oil bath for 10 min to allow sufficient time for foaming to happen. Saturated samples were foamed at increasingly higher foaming temperatures while keeping all other processing parameters the same. The temperature interval of the two adjacent foaming temperatures was 5 °C. After foaming, the sample was immediately quenched in an oil bath which was kept much colder than the foaming oil bath, to stop further foaming. The minimum foaming temperature was determined as the average value of the two adjacent foaming temperatures: at the lower temperature, sample remains transparent and at the higher temperature, sample becomes opaque. Opaqueness indicates the formation of bubbles. In addition, SEM images of these samples were prepared to confirm the formation of cells. Previous studies suggested that the minimum foaming temperature and T_g of polymer-diluent system can be considered to be equivalent [19,20].

3. Data analysis, results and discussions

3.1. Solubility

Fig. 1 shows the mass percentage (or mass%) of CO₂ in PMMA as a function of time at various saturation temperatures T_{sat} , ranging from -30 °C to 100 °C. The saturation pressure was fixed at 5 MPa. Solubility is defined as the equilibrium mass% of CO₂ at a specified condition, and usually expressed as a percentage of the original polymer mass, e.g. 10% CO₂ means that 10% of the mass of original PMMA is now absorbed into the PMMA. Solubility and equilibrium mass% of CO₂ are used interchangeably in this paper. Note that there was a short time interval between the pressure release and weight measurement in the sorption experiment. To account for the CO₂ lost during this interval, the solubility of CO₂ in the samples was obtained by extrapolating the desorption curve to zero desorption time. Detailed procedures can be found in literature [21,22]. Solubility results in a wider temperature range are summarized in Table 1. To better visualize the solubility trend, we've also plotted the solubility as a function of saturation temperature in Fig. 2. It is evident that the amount of CO₂ absorbed increases with decreasing temperature. By decreasing temperature from 100 °C to -30 °C, the solubility increases from 3.9% to 39.3%, a ten-fold increase. To our knowledge, 39.3% is the highest amount of CO₂ ever reported absorbed in PMMA. Compared to other polymers at similar pressure and temperature conditions [2,23,24], PMMA has

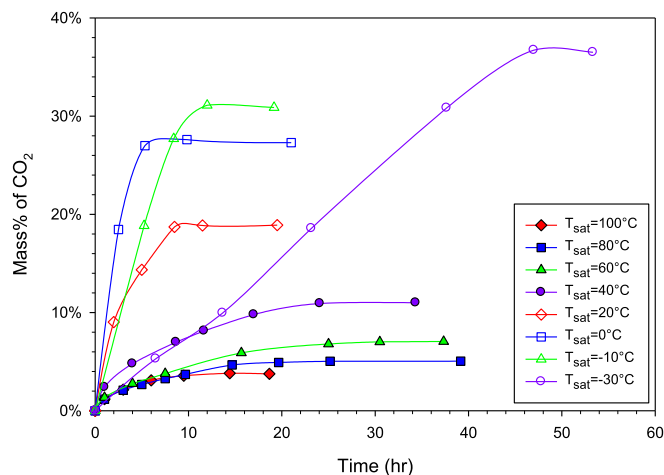


Fig. 1. Mass% of CO₂ in PMMA as a function of time at various saturation temperatures. Sample thickness was 1.48 mm.

Table 1

Summary of solubility (S), diffusivity (D), and saturation time (t_{sat}) at various saturation temperatures (T_{sat}).

T_{sat} (°C)	S	D (cm ² /s)	t_{sat} (hr) ^a
100	3.9%	1.41×10^{-7}	14
80	5.2%	7.8×10^{-8}	25
60	7.4%	5.8×10^{-8}	31
50	8.7%	5.5×10^{-8}	30
40	11.4%	6.6×10^{-8}	24
30	14.5%	7.5×10^{-8}	18
20	20.5%	1.8×10^{-7}	9
15	25.2%	2.78×10^{-7}	5
10	26.9%	3.65×10^{-7}	5
0	30.1%	2.63×10^{-7}	7
−10	32.6%	1.25×10^{-7}	12
−20	35.6%	5.5×10^{-8}	22
−30	39.3%	2.5×10^{-8}	47

^a for sample thickness 1.48 mm.

very high CO₂ solubility, which is probably due to the high mass density of carbonyl groups in PMMA [25].

The temperature dependence of solubility is typically given by Arrhenius equation [3,15],

$$S = S_0 \exp\left(-\frac{\Delta H_S}{RT}\right) \quad (1)$$

where S_0 is the pre-exponential factor, ΔH_S is the heat of sorption or enthalpy change upon solution of gas in the polymer, and R is the gas constant.

In Fig. 3, we plotted the natural logarithm of solubility as a function of the reciprocal of saturation temperature. Note the X-axis is $1000/T$ where T is in Kelvin. The plot shows two distinct regions for liquid and gaseous CO₂ that intersect at 15 °C, the phase change temperature for CO₂ at 5 MPa. Below and above this phase changing temperature, the data follows a linear trend: below 15 °C, ΔH_S is calculated to be −5.6 kJ/mol; above 15 °C, ΔH_S is −19.4 kJ/mol. Negative heat of sorption values indicate the exothermic nature of CO₂ sorption in PMMA. The difference between these two ΔH_S values is 13.8 kJ/mol. This value is close to the heat of condensation (or heat of vaporization) of CO₂, which is about 11.3 kJ/mol [26]. The change in heat of sorption at the phase changing temperature of CO₂ was also observed in our recent study of polycarbonate-CO₂ system [22].

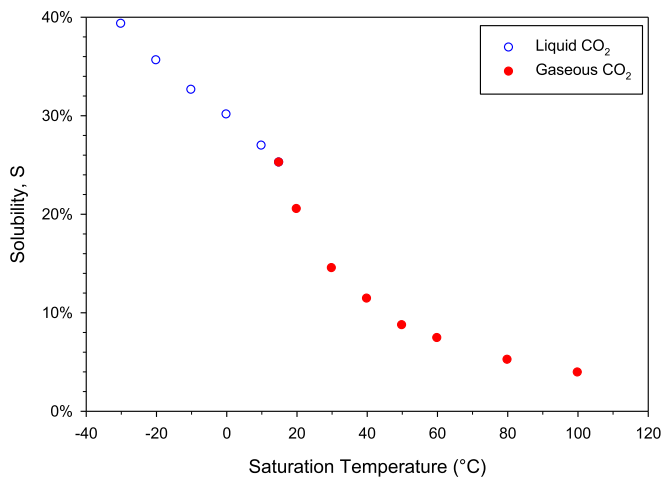


Fig. 2. Solubility of CO₂ in PMMA as a function of saturation temperature.

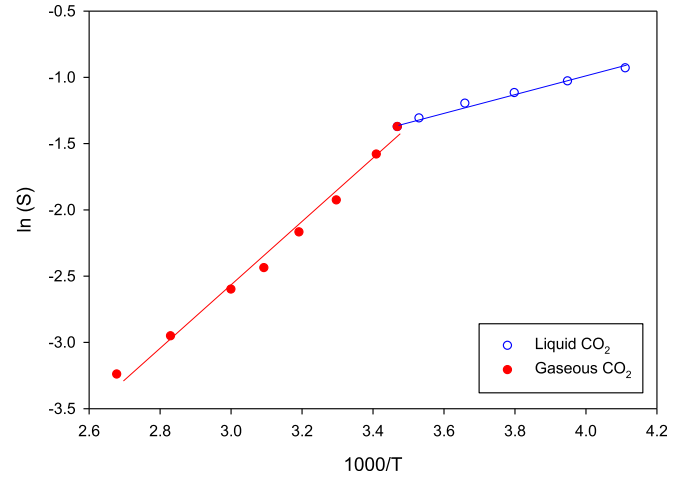


Fig. 3. Natural logarithm of solubility as a function of reciprocal of saturation temperature with best-fit lines. Note the change in slope when CO₂ changes from gas to liquid at 15 °C.

3.2. Diffusivity

Also, from Table 1, we can see that the time needed to reach equilibrium is different for different saturation temperatures, with −30 °C taking the longest (~47 h), and 10 °C & 15 °C the shortest (~5 h). Sorption diffusivity is usually used to characterize how fast the sorption takes place. One of the methods used to determine diffusivity from a sorption plot is the hybrid method. It is a more accurate method since the model fits the data from the whole sorption curve, including both short-time and long-time data [27]. The hybrid method is described by the following equations:

$$\frac{M_t}{M_0} = \phi(x)f(x) + [1 - \phi(x)]g(x) \quad (2)$$

$$x = Dt/L^2 \quad (3)$$

$$f(x) = 4(x/\pi)^{0.5} \quad (4)$$

$$g(x) = 1 - \left(8/\pi^2\right)\exp(-\pi^2x) \quad (5)$$

where M_t is the mass of CO₂ absorbed at time t , M_0 is the mass of CO₂ absorbed at equilibrium, $\phi(x)$ is the weighting function, $f(x)$ is the short-time approximation, $g(x)$ is the long-time approximation, D is the diffusivity, and L is the polymer thickness. The weighting function $\phi(x)$ is defined as:

$$\phi(x) = 1, \quad x \leq 0.05326$$

$$\phi(x) = 0, \quad x > 0.05326$$

Diffusivities are obtained by regression fitting the Eq. (2) to the sorption curves in Fig. 1. Fig. 4 shows an example of the model fitting to the data for sorption at 30 °C. The model predicts the sorption data quite well. Sorption diffusivities are summarized in Table 1. Also, we've plotted the diffusivity data in Fig. 5. We can see that temperature has a profound effect on the diffusivity in PMMA-CO₂ system. An order of magnitude variation can be seen, with the lowest diffusivity 2.5×10^{-8} cm²/s at −30 °C and highest 3.65×10^{-7} cm²/s at 10 °C. More importantly, from the figure, we can see the diffusivity

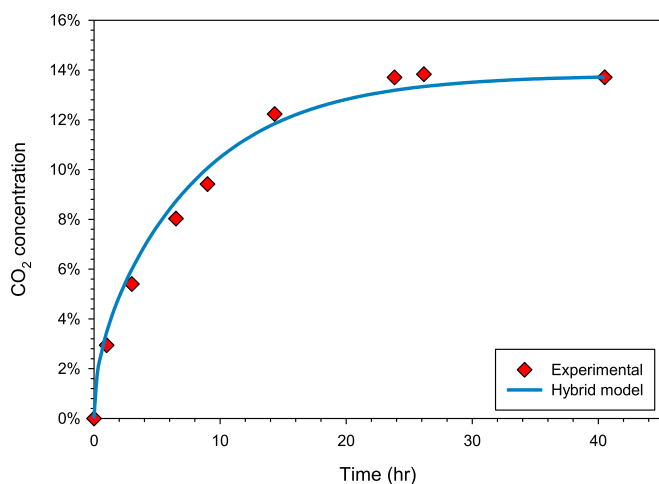


Fig. 4. Hybrid model fitting to the experimental data of CO₂ sorption in PMMA at 30 °C. D used is 7.5×10^{-8} cm²/s.

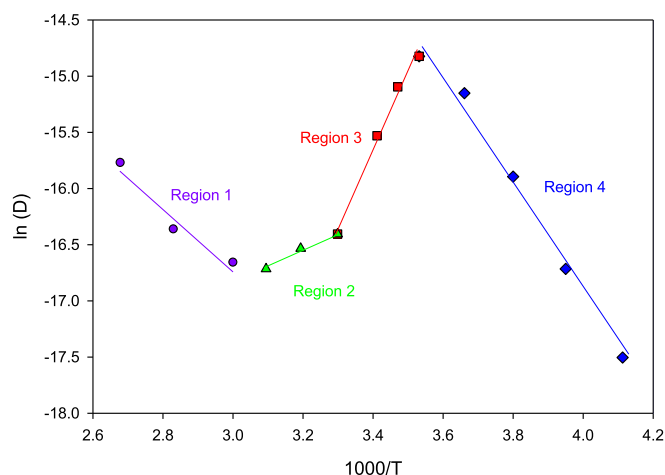


Fig. 6. Natural logarithm of diffusivity as a function of reciprocal of saturation temperature. Four distinct regions are labeled and best-fit lines for each region are also shown.

doesn't monotonically increase with increasing temperature over the entire temperature range, but instead, diffusivity increases with temperature in the range of -30 °C– 10 °C (region 4), decreases in the range of 10 °C– 30 °C (region 3), decrease more slowly in the range of 30 °C– 50 °C (region 2), and increases again in the range 60 °C– 100 °C (region 1). This is in contrast to many other polymer-gas systems, which typically show a monotonic increase of diffusivity with temperature.

The effect of temperature on the diffusivity of gas in polymers was first shown by Barrer [28] and later confirmed by others [3,15] to be that of an activated process obeying the Arrhenius relationship:

$$D = D_0 \exp\left(-\frac{\Delta H_D}{RT}\right) \quad (6)$$

where D_0 is the pre-exponential factor, ΔH_D is the activation energy for diffusion, R is gas constant and T is temperature in Kelvin.

In Fig. 6, we have plotted natural logarithm of diffusivities in Table 1 as a function of the reciprocal of saturation temperature. The four distinct regions mentioned before are also labeled. Temperatures range and physical state of PMMA-CO₂ in each region are summarized in Table 2.

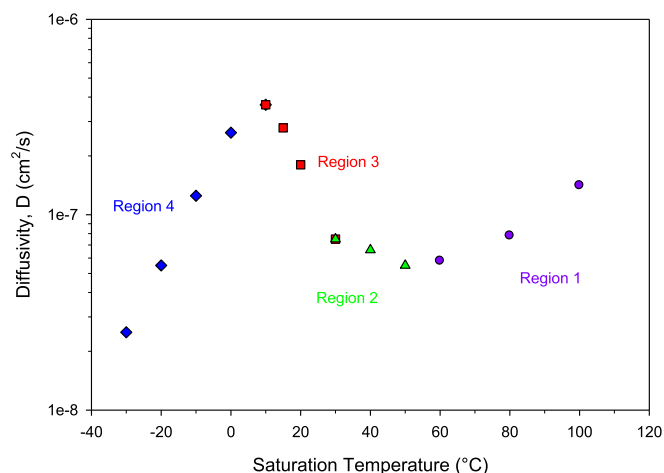


Fig. 5. Diffusivity of CO₂ in PMMA as a function of saturation temperature. Four distinct regions are identified where diffusivity varies differently with temperature.

Also, within each region, a best-fit line using Arrhenius equation is shown. In region 1, as the temperature decreases, the diffusivity decreases. An activation energy barrier ΔH_D of 22.8 kJ/mol needs to be overcome for CO₂ diffusion in region 1. In region 2, as the temperature decreases, the diffusivity increases. ΔH_D is calculated to be -12.6 kJ/mol. In region 3, as the temperature decreases, the diffusivity increases, at a much faster rate than in the region 2. ΔH_D for region 3 is calculated to be -57.6 kJ/mol. In region 4, as the temperature decreases, the diffusivity decreases, at a faster rate than in region 1. ΔH_D is calculated to be 39.8 kJ/mol.

Handa et al. [15] observed that at saturation pressure of 3.4 MPa, diffusion of CO₂ in PMMA at 0 °C was faster than that at 10 °C (lower diffusivity at a higher temperature), implying a negative activation energy. Negative activation energy has been reported in chemical reactions, where the rates of reaction decreased with increasing temperature [29,30]. The proposed mechanism was the formation of an intermediate complex. Regarding to diffusion, negative activation energies have only been reported in atom and ion diffusion [31,32]. Radiation-enhanced-diffusion rates for Ag in Ni have been found to decrease at elevated temperatures and negative activation energies were reported [31]. The proposed mechanism was that at elevated temperatures the concentration of long-lived complex defects (which enhanced the atom diffusion) decreased, resulting in a slower diffusion of Ag atoms in Ni. Also, it was reported that potassium ion release from zwitterionic phosphorylcholine containing pHEMA-based hydrogels was slower at higher temperatures [32]. Temperature-dependent swelling of hydrogels was proposed to explain the negative activation energy. Neither of these proposed mechanisms is applicable to our polymer-gas system. The physical mechanism for negative activation energy of diffusion in PMMA-CO₂ system is currently under investigation.

The state of the system in terms of glassy or rubbery described in Table 2 can be understood in terms of the plasticization of PMMA by the dissolved CO₂ at a given temperature. In the high temperature range (region 1), some amount of CO₂ is dissolved and

Table 2
Characteristics of the four distinct diffusion regions.

Region	Temperature range	Physical state of PMMA-CO ₂
Region 1	100 °C– 60 °C	Rubbery
Region 2	50 °C– 30 °C	Glassy
Region 3	30 °C– 10 °C	Rubbery
Region 4	10 °C to -30 °C	Glassy

plasticizes the polymer, and under the high saturation temperature the PMMA-CO₂ is in the rubbery state. As the temperature is lowered, the drop in the saturation temperature dominates the increased plasticization effect from slightly more CO₂ absorption, and the PMMA-CO₂ goes into the glassy state (region 2). As the temperature is further lowered, there is a rapid increase in CO₂ solubility, resulting in a larger plasticization effect. With increased plasticization, the PMMA-CO₂ goes into the rubbery state (region 3).

The transitions from region 1 to region 2 and region 2 to region 3 have been previously reported by Handa et al. using a similar diffusivity approach [15]. They showed that for PMMA-CO₂ at 5 MPa, the rubber-to-glass transition temperature were between 50 and 60 °C and the glass-to-rubber transition (retrograde vitrification) temperature between 20 and 30 °C. Our results on these two transition temperatures match well with their finding, with the rubber-to-glass transition temperature between 50 and 60 °C and the glass-to-rubber transition temperature at 30 °C.

A new transition from region 3 to region 4 is reported here. This transition occurs at around 10 °C, in the vicinity of the transition point (also the vaporization temperature of CO₂) in the solubility plot. As mentioned before, below the transition point, there is a change in heat of sorption ΔH_s , leading to a slower solubility increase with decreasing temperature (particularly evident from Fig. 2). If the ΔH_s were to stay the same when temperature is lowered, the solubility of CO₂ in PMMA would be much larger at low temperatures. Suppose we could continue the same solubility trend from above 10 °C (without the change in ΔH_s) to below 10 °C in the solubility plot, the solubility would be about 70% at –30 °C. But the measured solubility is only 39.3%. Without enough solubility and associated plasticization effect, the PMMA-CO₂ becomes glassy again.

As will be discussed in the next section, the existence of these four regions are reinforced with the comparison between glass transition temperature of PMMA-CO₂ mixture and corresponding saturation temperature used to achieve the equilibrium mass% of CO₂ in the mixture.

3.3. Glass transition temperature

Absorption of diluent into a polymer lowers its glass transition temperature, because the diluent molecules increase polymer intermolecular distance, decreasing intermolecular interactions. The weakening of these interactions increases the segmental mobility, which leads to a lower glass transition temperature. CO₂ has a high plasticization effect and can reduce T_g to a value significantly lower than that of the original polymer. This is one of the primary driving mechanisms in solid-state microcellular foaming. Therefore, knowledge of T_g of the polymer-diluent mixture is very important to understand and control the foaming process.

Fig. 7 shows the glass transition temperature (T_g) of PMMA-CO₂ as a function of equilibrium mass% of CO₂. The T_g reduces from 103 °C to –12.5 °C when 39.3% CO₂ is incorporated into PMMA. Handa et al. previously reported T_g of PMMA-CO₂ mixture up to 23.5% CO₂ [15]. The data are included in the plot. Both sets of data agree well for the equilibrium mass% of CO₂ between 7.5% and 23.5%. Data from our study cover a much larger range of mass% of CO₂. In PMMA, the T_g reduces from 103 °C to –12.5 °C with 39.3% CO₂, an average of 3 °C reduction with 1% increase in the amount of CO₂ absorbed. This plasticization effect is small in comparison to other polymers. For example, with 1% increase in CO₂ absorption, the reduction of T_g in polyethersulfone (PES), polysulfone (PSU), and cyclic olefin copolymer (COC), is 13 °C, 16 °C and 18 °C, respectively [20].

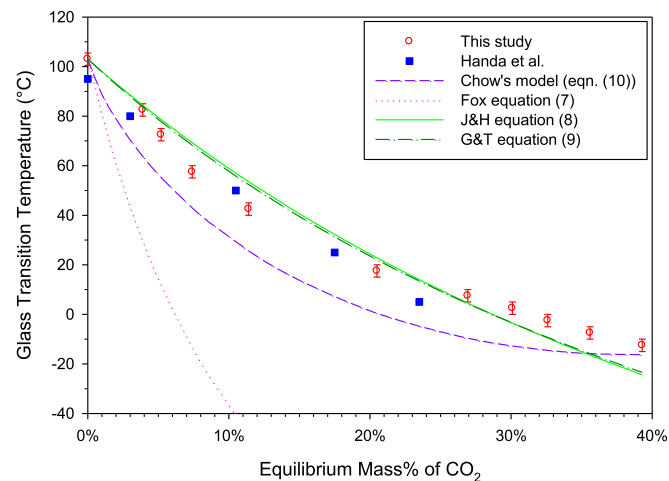


Fig. 7. T_g of PMMA-CO₂ mixture as a function of equilibrium mass% of CO₂. Experimental data from this study and the study by Handa et al. [15], and predictions using Fox equation, Jenckel & Heusch (J&H) equation, Gordon & Taylor (G&T) equation and Chow's model with $z = 2$ are shown.

Empirical models have been developed to predict the T_g of a variety of binary mixtures, including the plasticizers in high polymers. Three of these models were used to fit our data - Fox equation [33], Jenckel & Heusche equation [34], and Gordon & Taylor equation [35]:

$$\text{Fox equation : } \frac{1}{T_g(w)} = \frac{w}{T_{g1}} + \frac{1-w}{T_{g2}} \quad (7)$$

$$\begin{aligned} \text{Jenckel \& Heusche equation : } T_g(w) \\ = wT_{g1} + (1-w)T_{g2} + w(1-w)k \end{aligned} \quad (8)$$

$$\text{Gordon \& Taylor equation } T_g(w) = \frac{wT_{g1} + \mu(1-w)T_{g2}}{w + \mu(1-w)} \quad (9)$$

T_{g1} and T_{g2} are glass transition temperatures of CO₂ and PMMA, respectively; k and μ are constants selected for a best fit. T_g of CO₂ is recently reported as –223 °C (50 K) [36] and T_g of PMMA is 103 °C.

Predictions from these equations are shown in Fig. 7. The Fox equation predicts the T_g very poorly. The best-fit constants selected are $k = -175$ and $\mu = 0.62$ for Jenckel & Heusche and Gordon & Taylor equation, respectively. Predictions from these two equations are very close to each other. Among various models selected, these two models can best fit our experimental data. However, they underestimate the T_g when below 24% equilibrium mass% of CO₂ and overestimate the T_g when above 24% equilibrium mass% of CO₂.

A model based on both classical and statistical thermodynamics predicting the T_g of diluent laden polymer was developed by Chow [37]. The model was successfully applied to predict the T_g of polystyrene with a variety of diluents and a range of mass% of diluents. The model is as follows:

$$\ln\left(\frac{T_g}{T_{g0}}\right) = \beta[(1-\theta)\ln(1-\theta) + \theta\ln\theta] \quad (10)$$

$$\beta = \frac{zR}{M_p \Delta C_{pp}} \quad (11)$$

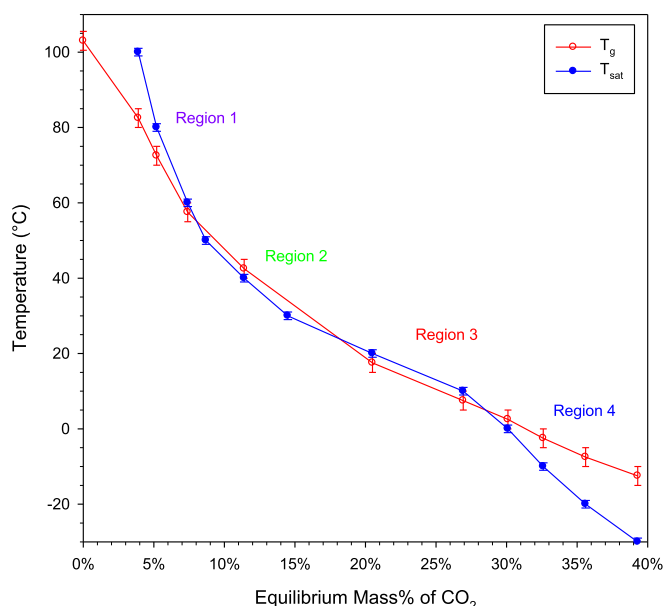


Fig. 8. Comparison between the T_g of PMMA–CO₂ mixture at each equilibrium mass% of CO₂ and the T_{sat} used to achieve the corresponding amount of CO₂. The four regions are also labeled.

$$\theta = \frac{M_p}{zM_d} \frac{w}{1-w} \quad (12)$$

where T_g and T_{g0} are the glass transition temperature for the polymer–diluent system and the neat polymer, respectively, M_p is the molar mass of the polymer repeat unit, M_d is the molar mass of diluent, R is the gas constant, w is the mass (or weight) fraction of the diluent, ΔC_{pp} is the excess transition isobaric specific heat of the polymer, and z is the lattice coordination number.

In the figure, we also show the T_g prediction based on Chow's model with $z = 2$ and $\Delta C_{pp} = 0.265 \text{ J K}^{-1} \text{ g}^{-1}$, which were suggested [37]. The model can predict the general decreasing trend of T_g ; however, it predicts the T_g values poorly with the prediction always lower than our experimental data.

Fig. 8 shows a comparison between the T_g of PMMA–CO₂ mixture at each equilibrium mass% of CO₂ and the saturation temperature (T_{sat}) used to achieve the corresponding amount of CO₂. The same four regions mentioned previously are also labeled. The relative magnitude of T_g and T_{sat} determines whether the PMMA–CO₂ mixture is in a glassy or rubbery state. In region 1 and region 3, at the equilibrium mass% of CO₂, T_{sat} is greater than T_g so the PMMA–CO₂ system is in the rubbery state. In region 2 and region 4, T_{sat} is below the T_g so the PMMA–CO₂ system is in the glassy state. The transition points between different regions are between 50 °C and 60 °C for region 1 → region 2, between 20 °C and 30 °C for region 2 → region 3 and between 0 °C and 10 °C for region 3 → region 4. These transition points match relatively well with the ones identified from diffusivity data (see Table 2). At very low T_{sat} and thus high equilibrium mass% of CO₂, two lines are diverging from each other. The decrease in T_{sat} exceeds the reduction of T_g with the absorbed CO₂, so the system is in the glassy state.

4. Summary

The effect of saturation temperature (T_{sat}) on sorption and diffusion of CO₂ in PMMA was investigated over the range –30 °C to 100 °C. Saturation temperature has significant effects on both the

solubility and diffusivity. Solubility increases with decreasing saturation temperature, a ten-fold increase from 3.9% at 100 °C to 39.3% at –30 °C. A change in heat of sorption has been found to occur around 15 °C, the vaporization temperature at 5 MPa for CO₂. The change in heat of sorption matches with heat of vaporization of CO₂. The effect of temperature on solubility can be described with an Arrhenius relationship.

For diffusivity, four distinct regions are identified, where diffusivity may increase or decrease with decreasing temperature, characterized by the different activation energies. As T_{sat} is lowered from 100 °C to –30 °C, a rubber-to-glass transition between 50 and 60 °C and a glass-to-rubbery transition at 30 °C have been observed, which match with previously reported results. In addition, we observed a third transition from rubbery state to glassy state at 10 °C, in the vicinity of CO₂ vaporization temperature at 5 MPa (15 °C).

Dissolution of CO₂ into PMMA plasticizes the polymer. Thus, at T_{sat} of –30 °C, incorporation of 39.3% CO₂ in PMMA decreases the T_g from 103 °C to –12.5 °C. The minimum foaming temperature, or equivalently the effective T_g of the polymer–gas mixture, shows a non-linear decreasing trend with the amount of CO₂ absorbed.

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